

IMDb Cinema & TV Evolution Explorer (1920–2025)

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GitHub Repository: github.com/b14ckPanther/imdb-cinema-tv-evolution

Live Project: [IMDb Cinema & TV Evolution Explorer](#)

1. Introduction

Cinema and television have changed considerably over the past century in terms of production volume, genre composition, audience reception, and popularity. While IMDb provides extensive information about individual titles, examining these records separately makes it difficult to identify broader patterns and compare developments across different periods.

This project presents an interactive visualization of IMDb data for feature films and television series released between 1920 and 2025. It enables users to examine historical trends in release volume, audience ratings, popularity, runtime, and genre, while also comparing the characteristics of movies and TV series.

Research Question:

How have cinema and television evolved from 1920 to 2025 in terms of production volume, audience ratings, popularity, runtime, genre composition, and the differences between feature films and TV series?

The project is intended for users interested in exploring historical trends in cinema and television, including students, researchers, and film and television enthusiasts. Interactive filtering and coordinated visualizations allow the data to be examined at different levels, from long-term historical patterns to specific periods, genres, and formats.

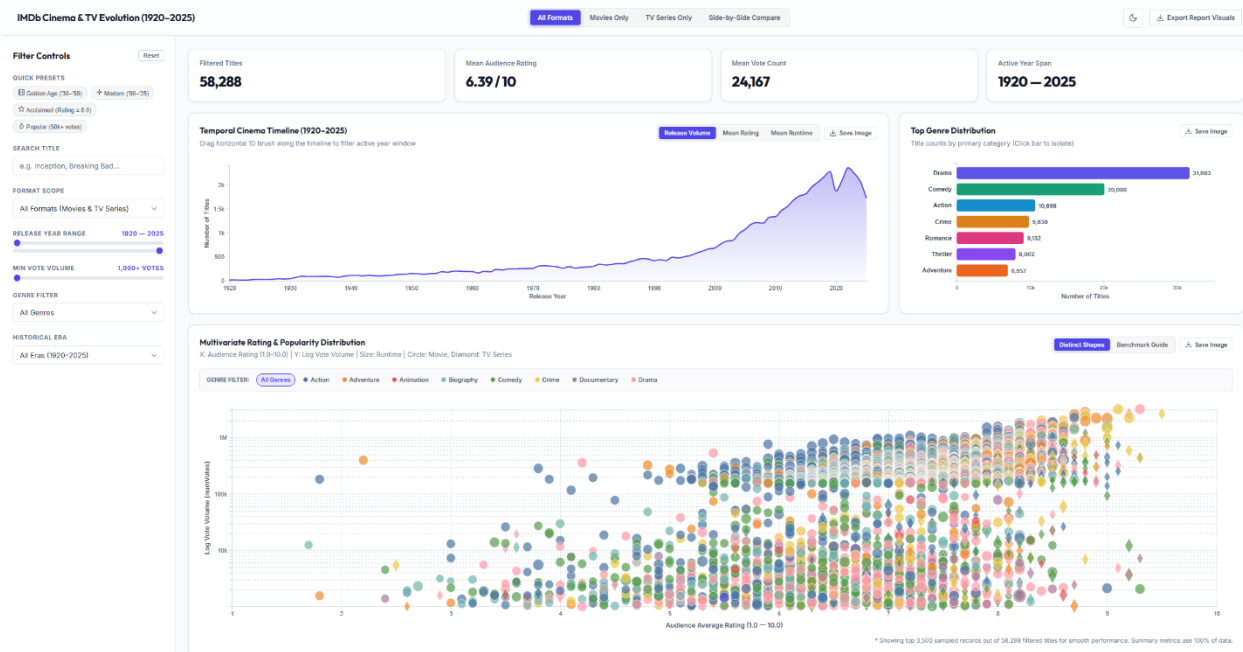


Figure 1. Overview of the IMDb Cinema & TV Evolution Explorer, showing the main visualizations, summary measures, and interactive filtering controls.

2. Dataset Description

The project uses the **IMDb Non-Commercial Datasets**, combining title metadata and audience rating information through IMDb's unique title identifier, tconst. The two IMDb source files used in the project are:

- [title.basics.tsv.gz](#) – contains title type, title name, release year, runtime, and genres.
- [title.ratings.tsv.gz](#) – contains the average IMDb rating and number of user votes for each rated title.

The source datasets are substantially larger than the final analytical subset. **title.basics.tsv** contains approximately 12.67 million title records, while **title.ratings.tsv** contains approximately 1.70 million rated-title records. To create a focused dataset suitable for the research question and interactive visualization, the analysis was restricted to feature films and TV series released between 1920 and 2025 with at least 1,000 IMDb votes. After preprocessing, the final analytical dataset contains 58,288 titles.

Table 1. Main attributes used in the visualization

Attribute	Description	Use in Visualization
tconst	Unique IMDb title identifier	Identifying individual titles
primaryTitle	Primary title name	Search, tooltips, and title details
titleType	Movie or TV series	Format filtering and comparison
startYear	Release or series start year	Timeline and year filtering
averageRating	IMDb audience rating (1–10)	Rating analysis and scatterplot
numVotes	Number of IMDb user votes	Popularity measure
runtimeMinutes	Runtime in minutes	Runtime analysis and point size
genres	Genres associated with the title	Genre filtering and distribution

Dataset Source: [IMDb Non-Commercial Datasets](#)

3. Data Preparation

The raw IMDb files required preprocessing before they could be used efficiently in the interactive visualization. Processing was performed in **Python using Pandas and PyArrow**, with large files processed in chunks to reduce memory requirements.

The main preprocessing steps were:

1. **Title selection:** Only records classified as movie or tvSeries were retained.
2. **Time range:** Only titles from 1920–2025 with a valid start year were included.
3. **Ratings integration:** Title metadata was joined with the ratings dataset using the unique tconst identifier.
4. **Vote threshold:** Titles with fewer than 1,000 IMDb votes were excluded to focus the analysis on titles with a meaningful level of audience participation.
5. **Data cleaning:** Missing and invalid values required for the analysis were handled, and implausible runtime values were excluded from runtime-based calculations.
6. **Genre preparation:** Genre information was prepared for consistent filtering and aggregation.
7. **Web optimization:** The processed data was exported as compact JSON files for efficient loading and interaction in the browser.

The resulting dataset of 58,288 titles serves as the common data source for all visualizations and interactive analyses presented in the project.

3.1 Tools and Technologies

The project was implemented as a web-based interactive visualization using Vite, D3.js, Vanilla JavaScript (ES Modules), and Vanilla CSS3. D3.js was used to construct the interactive charts and visual encodings, while Vite was used for development and production builds. Python, Pandas, PyArrow, and NumPy were used for data preprocessing and preparation of the optimized JSON files used by the application. The source code is managed through GitHub, and the deployed application is hosted on Vercel.

ChatGPT and Antigravity were used as AI-assisted development tools. ChatGPT supported research planning, visualization decisions, interpretation of results, and documentation. Antigravity was used as a multi-agent development environment to support data auditing, preprocessing, visualization implementation, debugging, and build verification.

4. Visualization Design

The visualization was designed as a coordinated dashboard in which multiple views present complementary aspects of the same dataset. User interactions—including filtering by time period, format, genre, vote threshold, historical era, and title search—are reflected across the visualizations. This coordination allows users to move between an overview of long-term historical trends and more focused comparisons of specific subsets of the data.

The dashboard contains three main visualizations:

Temporal Cinema Timeline. A time-series area chart shows changes from **1920 to 2025**. Users can switch between release volume, mean audience rating, and mean runtime. This representation supports the identification of long-term trends and changes across historical periods.

Top Genre Distribution. A horizontal bar chart ranks the most frequent genres in the current selection. Bar length represents the number of titles, providing a direct comparison of genre prevalence across different periods and formats.

Multivariate Rating & Popularity Distribution. A scatterplot combines several attributes at the title level. Audience rating is mapped to the horizontal axis and vote count to a logarithmic vertical axis, which accommodates the large differences in popularity between titles. Color represents genre, point size represents runtime, and when the distinct-shapes option is enabled, circles represent movies and diamonds represent TV series. Together, these encodings support the simultaneous examination of rating, popularity, runtime, genre, and format.

IMDb Multivariate Rating & Popularity Distribution
IMDb Cinema & TV Evolution Explorer • Exported 8/19/2026

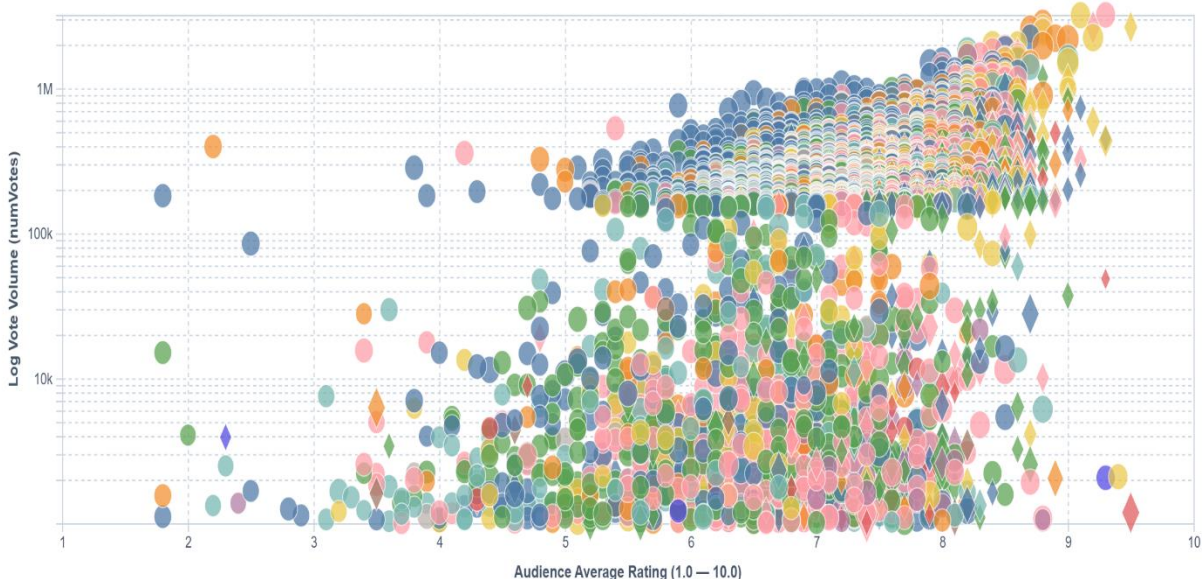


Figure 2. *Multivariate Rating & Popularity Distribution.* Audience rating is shown on the horizontal axis and vote volume on a logarithmic vertical axis. Color represents genre, point size represents runtime, and, with distinct shapes enabled, circles and diamonds distinguish movies from TV series.

5. Interaction and Exploration

Interactivity allows users to examine the dataset at different levels rather than relying on a fixed view of the data. The dashboard provides controls for release-year range, minimum vote count, genre, historical era, title search, and format. Quick presets provide access to selected periods and commonly used subsets, while the format controls allow users to explore all formats, movies only, TV series only, or a side-by-side comparison.

The visualizations also support direct interaction. Users can brush across the timeline to select a time period, select a genre from the distribution chart, and hover over individual titles in the scatterplot to view additional information. Title details are available on demand, keeping the main visualization readable while allowing individual records to be examined when needed.

The summary measures at the top of the dashboard update according to the current selection and display the number of filtered titles, mean audience rating, mean vote count, and active year span. These measures provide numerical context for the patterns shown in the visualizations.

Coordinated Filtering

The dashboard views are coordinated through a shared application state. Changes to the selected period, format, genre, or minimum vote threshold are reflected across the relevant visualizations and summary measures. Multiple filters can therefore be combined to move from an overall historical view to more focused subsets while preserving a consistent visual structure.

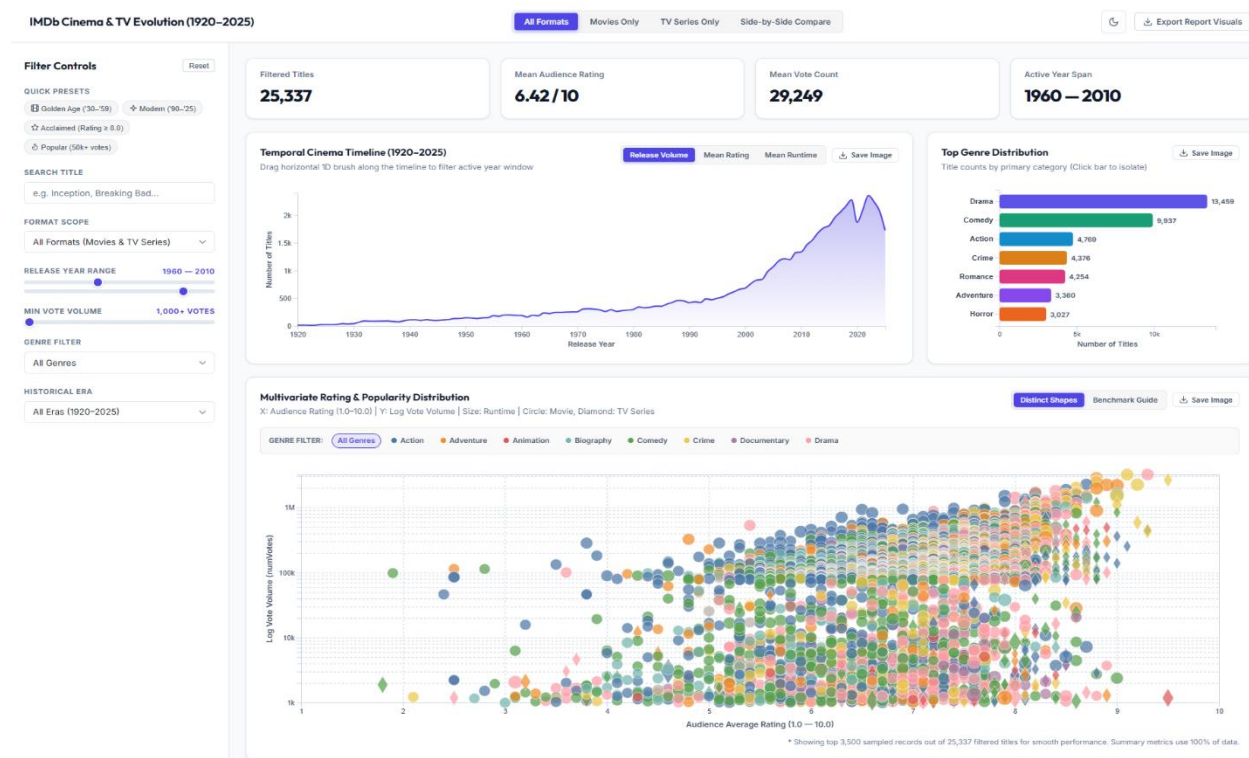


Figure 3. Example of coordinated filtering with the active year range restricted to 1960–2010. The summary measures and coordinated visualizations update together to reflect the selected subset.

6. Analysis and Findings

6.1 Evolution of Production Volume

The clearest long-term change in the dataset is the growth in the number of IMDb titles released over time. Figure 4 shows the annual release volume for the 58,288 movies and TV series included in the analysis. Release activity remains relatively low during the early decades of the dataset and increases gradually through much of the twentieth century. The rate of growth becomes substantially stronger from the 1990s onward, with the highest annual release volumes occurring during the most recent decades.

The timeline also shows that this growth was not continuous. A visible decline occurs around 2020, followed by a recovery and another high point in the following years. Release volume then decreases toward the end of the observed period. Despite these fluctuations, the dominant long-term pattern is a substantial expansion in the number of productions represented in the dataset.

IMDb Temporal Cinema Timeline (1920–2025)
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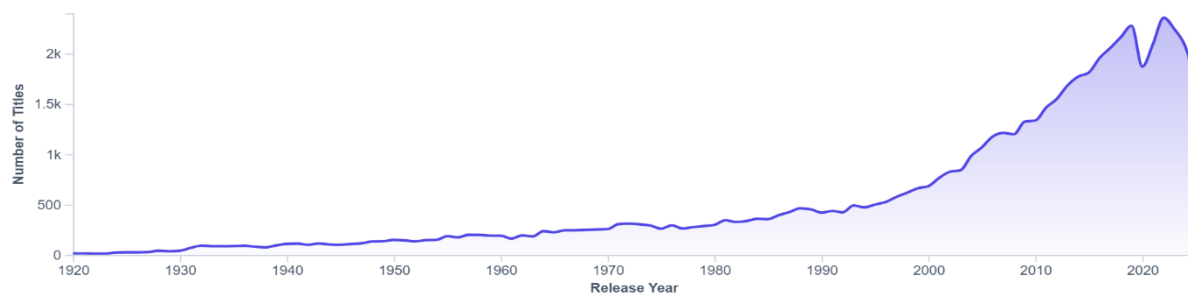


Figure 4. Annual release volume of movies and TV series in the analytical dataset from 1920 to 2025. The timeline shows long-term growth in the number of qualifying titles, with particularly strong expansion in recent decades.

6.2 Changes in Audience Ratings

Production volume changed considerably across the period, but audience ratings followed a different pattern. Figure 5 presents the mean IMDb audience rating by year. Unlike release volume, which increases strongly over time, average ratings remain within a comparatively narrow range.

The earlier decades contain several relatively high annual averages, while the later decades show somewhat lower and more stable yearly averages. This pattern should not be interpreted as direct evidence that newer productions are of lower quality. The number and composition of titles differ considerably between periods, and the dataset includes only titles that received at least 1,000 IMDb votes. The visualization therefore describes changes in the ratings of the titles represented in the analytical dataset rather than an objective measure of changes in production quality.

IMDb Temporal Cinema Timeline (1920–2025)
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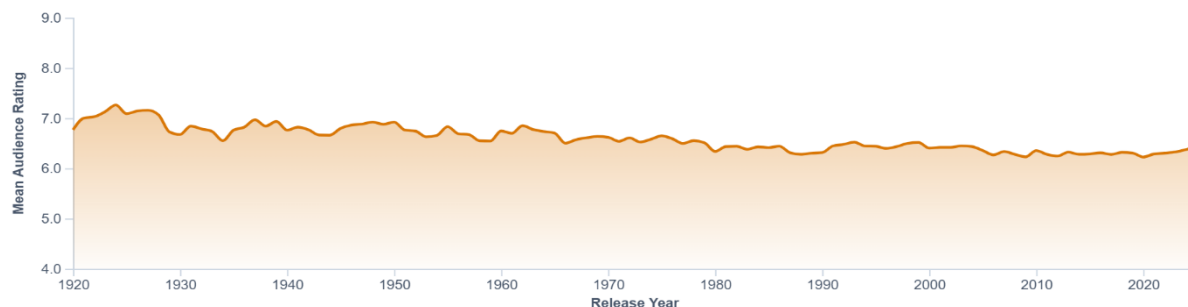


Figure 5. Mean IMDb audience rating by year from 1920 to 2025. Compared with the strong increase in release volume, annual mean ratings vary within a considerably narrower range.

6.3 Genre Distribution

Genre composition provides another perspective on the structure of the analytical dataset. Figure 6 shows the most frequent genres among the 58,288 titles included in the analysis. Drama is the largest category, followed by Comedy, while Action, Crime, Romance, Thriller, and Adventure also account for substantial numbers of titles.

The differences between categories are considerable. Drama appears in 31,693 titles and Comedy in 20,098, compared with 10,698 for Action and fewer than 10,000 for each of the remaining genres shown. This indicates that the dataset is not evenly distributed across genres and that Drama and Comedy represent particularly large portions of the titles that satisfy the project's selection criteria.

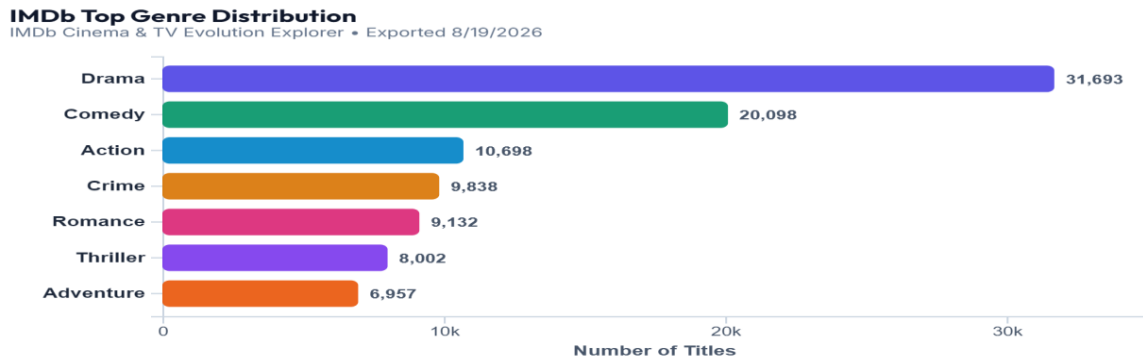


Figure 6. *Distribution of the most frequent genres in the analytical dataset. Drama is the most common genre, followed by Comedy and Action.*

6.4 Relationship Between Audience Rating and Popularity

Figure 7 examines the relationship between IMDb audience rating and popularity, measured by the number of IMDb votes. Each point represents a title, with vote count displayed on a logarithmic scale because popularity varies substantially across the dataset.

The visualization shows that titles with similar audience ratings can have very different levels of popularity. This is especially visible within the main concentration of titles between approximately 6 and 8 rating points, where vote counts extend across a wide range. Highly rated titles are therefore not necessarily the most widely voted-on titles, and titles with comparable ratings may differ considerably in audience reach.

The additional visual encodings provide further context: color distinguishes genre, point size represents runtime, and distinct shapes differentiate movies from TV series when enabled. The scatterplot therefore supports comparison across several attributes while preserving the central distinction between audience evaluation and popularity.

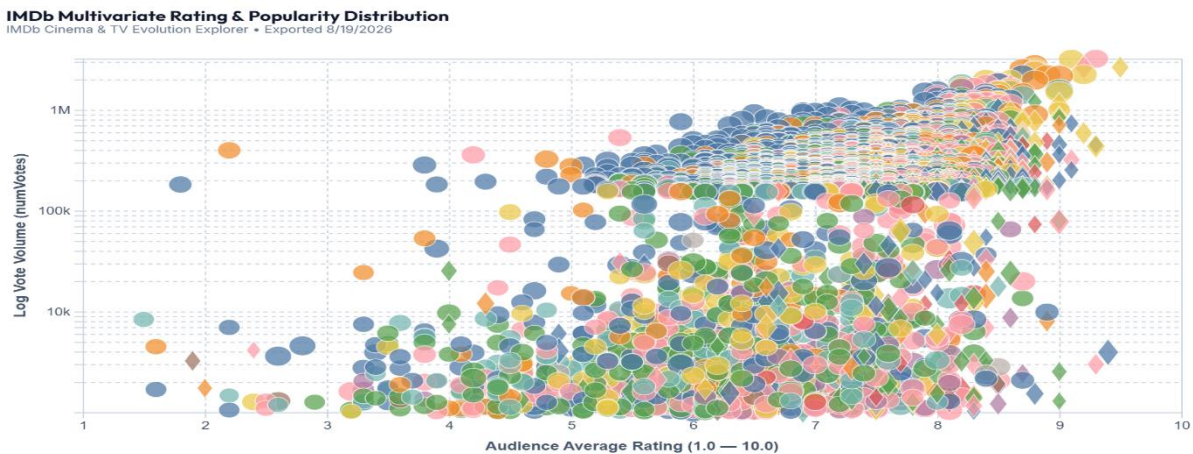


Figure 7. *Relationship between IMDb audience rating and vote volume. Vote count is shown on a logarithmic scale; color represents genre, point size represents runtime, and distinct shapes differentiate movies from TV series.*

6.5 Movies and TV Series

The dashboard also supports comparison between movies and TV series. In the standard view, both formats are displayed within the same rating–popularity space, with circles representing movies and diamonds representing TV series when distinct shapes are enabled. The Side-by-Side Compare mode provides a more focused comparison by changing the scatterplot arrangement while preserving the same axes and visual encodings.

Figure 8 shows this comparison across the full 1920–2025 period. Both formats occupy overlapping ranges of audience rating and vote volume, but their distributions can be examined separately without changing the underlying measures. Keeping the same rating and popularity scales makes the comparison consistent and allows differences in the concentration and spread of the two formats to be identified more clearly.

IMDb Multivariate Rating & Popularity Distribution
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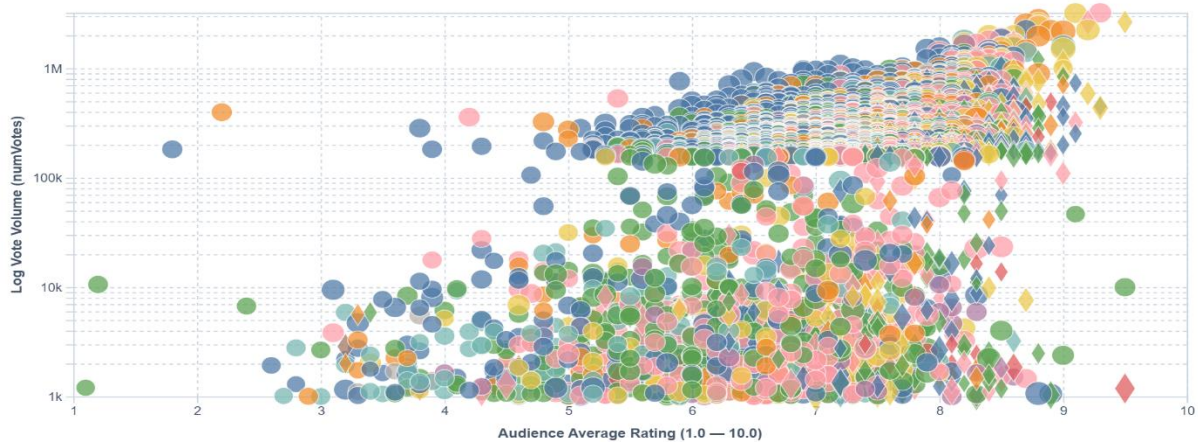


Figure 8. Movie and TV-series comparison using Side-by-Side Compare mode. Audience rating and vote volume retain the same scales, while the comparison layout separates the two formats more clearly within the multivariate visualization.

6.6 Comparison Across Historical Periods

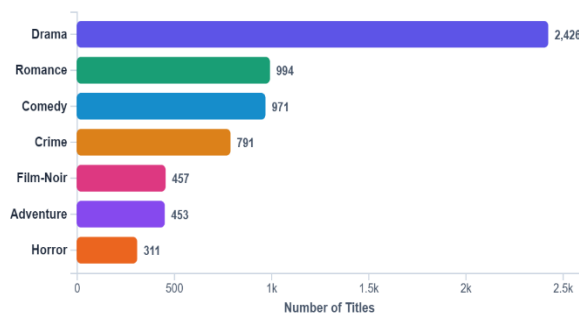
The **Golden Age (1930–1959)** and **Modern Period (1990–2025)** were compared using the same minimum threshold of 1,000 IMDb votes. The selected subsets differ substantially in size: 3,624 titles are included in the Golden Age compared with 45,626 in the Modern Period. Their mean audience ratings are 6.75 and 6.33, respectively.

Figure 9 also shows changes in genre composition. Drama is the most frequent genre in both periods, while the remaining rankings differ. Romance and Comedy follow Drama in the Golden Age, with Film-Noir also among the leading genres. In the Modern Period, Comedy and Action follow Drama, while Thriller appears among the leading categories. These results describe differences within the selected IMDb titles and should not be interpreted as a direct measure of changes in production quality.

(a) Golden Age (1930–1959)

IMDb Top Genre Distribution

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(b) Modern Period (1990–2025)

IMDb Top Genre Distribution

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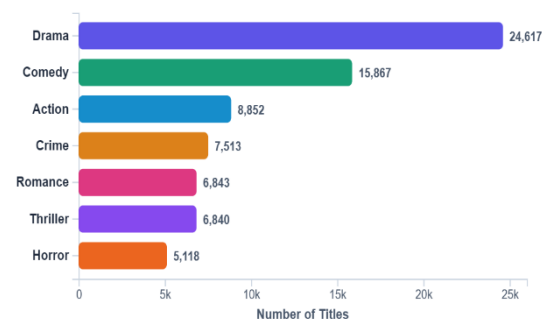


Figure 9. Comparison of the most frequent genres in (a) the Golden Age (1930–1959) and (b) the Modern Period (1990–2025), using the same minimum threshold of 1,000 IMDb votes.

6.7 Interactive Filtering and Focused Exploration

The dashboard allows multiple filters to be combined in order to examine more specific subsets of the analytical dataset. Filters can be applied by release year, minimum vote count, genre, format, and historical period. The summary measures and applicable visualizations update according to the active selection, allowing the same visual structure to be used for both broad exploration and focused comparisons.

One example is the Popular preset, which raises the minimum vote threshold from 1,000 to 50,000 IMDb votes. This reduces the selected dataset from 58,288 to 5,203 titles and shifts the analysis toward productions that received substantially greater audience attention. The resulting view can be used to examine whether the genre distribution, audience-rating range, and movie–TV composition of highly voted titles differ from those of the broader analytical dataset.

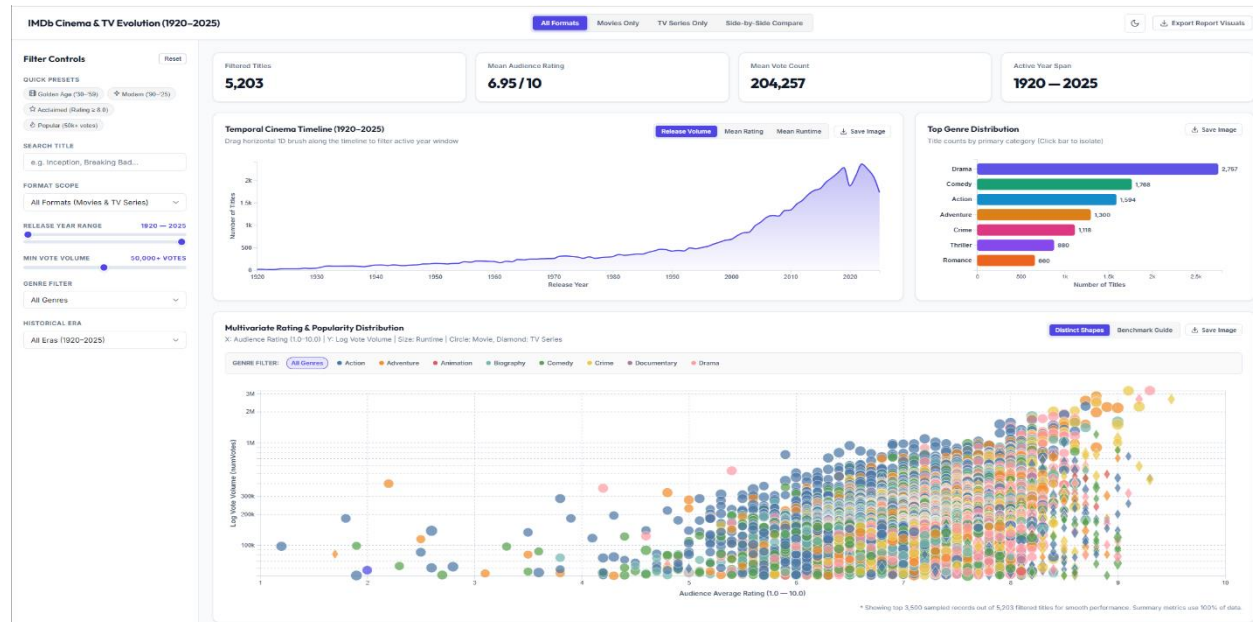


Figure 10. Dashboard after applying the Popular preset (50,000+ IMDb votes). The coordinated view contains 5,203 titles and updates the summary measures, genre distribution, and rating–popularity visualization to represent the more highly voted subset.

6.8 Effect of Popularity Threshold

Increasing the minimum vote threshold substantially reduces the number of titles included in the analysis and concentrates the visualization on productions with greater audience reach. This is particularly visible in the rating–popularity scatterplot, where titles with relatively low vote counts are removed while highly voted titles remain.

The threshold therefore serves both an analytical and a practical purpose. Analytically, it enables comparisons between broadly rated titles and more widely recognized productions. From a visualization perspective, the threshold provides a focused view of titles that attracted substantially greater audience participation. However, a higher threshold represents a more selective subset of IMDb and should not be interpreted as representative of all films and TV series.

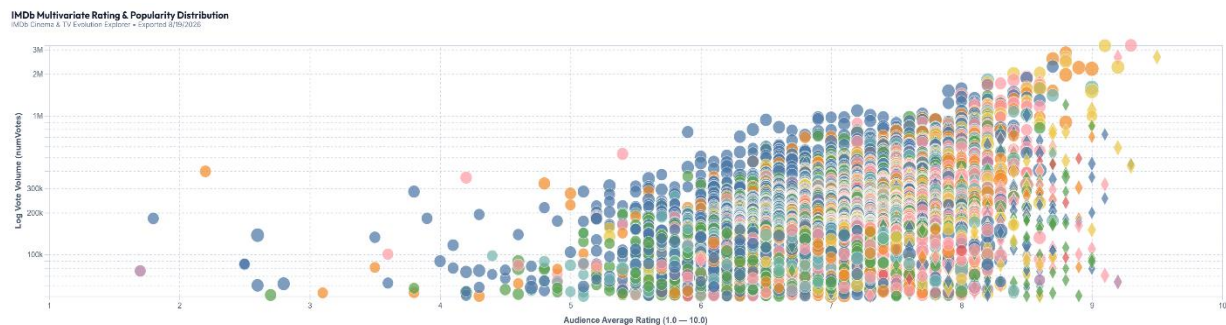


Figure 11. Rating and popularity distribution for titles with at least 50,000 IMDb votes. Raising the vote threshold focuses the analysis on titles with greater audience participation while preserving the same rating, genre, runtime, and format encodings.

7. Discussion

The analysis reveals several broad patterns in the IMDb titles included in the analytical dataset. The most prominent is the substantial growth in release volume over time, particularly in recent decades, while mean audience ratings remain within a comparatively narrow range. Genre composition is also uneven: Drama and Comedy are the most frequent genres overall, although the historical comparison shows that the relative prominence of other genres changes between periods.

The rating–popularity analysis indicates that audience rating and audience reach represent different dimensions. Titles with similar ratings can have substantially different vote counts, so a high IMDb rating does not necessarily correspond to high popularity. Movies and TV series also occupy overlapping regions of the rating–popularity space rather than forming completely separate groups.

Interactive filtering supports more focused examination of these patterns. Changing the period, format, genre, or vote threshold allows the same visualizations to be used for different subsets. For example, increasing the minimum vote threshold from 1,000 to 50,000 reduces the selected dataset from 58,288 to 5,203 titles and focuses the analysis on productions with greater audience participation.

7.1 Limitations

The results should be interpreted within the scope of the analytical dataset. Only movies and TV series released between 1920 and 2025 with at least 1,000 IMDb votes are included. The findings therefore describe this selected IMDb subset rather than all film and television productions.

IMDb ratings and vote counts reflect activity among IMDb users and should not be interpreted as objective measures of production quality or as representative of all viewers. Vote count is used in this project as an indicator of audience participation and popularity within IMDb.

The scatterplot also uses sampling for large filtered subsets to maintain responsive browser performance. Summary measures use the full filtered dataset, while the scatterplot displays up to 3,500 sampled titles. It is therefore intended primarily for examining broad distributions and patterns.

7.2 Design Trade-offs

The dashboard balances analytical detail with readability and performance. The multivariate scatterplot combines rating, vote count, runtime, genre, and format in one view, which supports multidimensional analysis but can also create visual density. Filtering and comparison controls help narrow the display when necessary.

Limiting the scatterplot to 3,500 displayed titles reduces rendering cost while the summary measures continue to represent the complete filtered subset. This maintains interactive performance while preserving the broader statistical context provided by the full-data summary measures.

8. Conclusions

This project examined the evolution of cinema and television from 1920 to 2025 using an interactive visualization based on IMDb data. The analysis combined temporal trends, audience ratings, vote counts, runtime, genre, and format within a coordinated dashboard, allowing these dimensions to be examined together and across different subsets of the data.

The results show that the most pronounced long-term change is the substantial increase in the number of represented titles, particularly in recent decades, while annual mean audience ratings remain within a considerably narrower range. Genre composition is uneven and varies across historical periods, and the analysis of rating and popularity demonstrates that highly rated titles are not necessarily the most widely voted. Movies and TV series also occupy overlapping regions of the rating–popularity space, although their distributions can be examined more clearly through the dedicated comparison mode.

Interactive filtering makes it possible to move beyond these overall patterns and examine specific periods, formats, genres, and popularity levels while maintaining the same visual framework. The project therefore demonstrates how coordinated interactive visualization can support the exploration of a large historical dataset and reveal patterns that are difficult to identify when individual IMDb records are considered separately.

9. References

IMDb. *IMDb Non-Commercial Datasets*.

<https://developer.imdb.com/non-commercial-datasets/>

IMDb. *title.basics.tsv.gz*.

<https://datasets.imdbws.com/title.basics.tsv.gz>

IMDb. *title.ratings.tsv.gz*.

<https://datasets.imdbws.com/title.ratings.tsv.gz>

10. Project Resources

Live Visualization: <https://imdb-cinema-tv-evolution.vercel.app/>

GitHub Repository: <https://github.com/b14ckPanther/imdb-cinema-tv-evolution>